Using Spatial Lag, Spatial Error, and Geographically Weighted Regression to Predict Median House Values in Philadelphia Block Groups

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Introduction

This report examines the relationship between median house values and several neighborhood characteristics across Philadelphia's census block groups. The ability to predict median house values provides insight into the housing market for politicians, developers, and city planners who are working to reshape the city. However, it is not always clear how to accomplish this due to many factors determining property values. In a previous report, this group conducted an ordinary least squares regression (OLS regression) with logarithmic-transformed median household values as a dependent variable. This report aims to address the limitations of this previous report by considering spatial autocorrelation.

Previously, this group found that an OLS regression, with independent variables being the percentage of vacant, the rate of single-house units, the percentage of individuals with a bachelor's degree, and the logarithmic-transformed number of households living in poverty, violates the assumption of independence among residuals due to spatial autocorrelation. Overall, OLS regressions are generally not appropriate for spatial data. Therefore, this report assesses the same question using three alternatives -- Spatial Lag, Spatial Error, and Geographically Weighted Regression (GWR) – to see whether these methods perform better than OLS and which are best performing.

Methods

Spatial Autocorrelation

Recalling Tobler's First Law of Geography, "Everything is related to everything else, but near things are more related to distant things", spatial correlation is the extent that a data point is correlated with itself across space. Visually, spatial autocorrelation reflects how values across a dataset or geography are spatially clustered or dispersed.

Global Moran's I

Moran's I (1950) is a widely used method of testing for spatial autocorrelation or spatial dependences. The formula for Moran's I is:

$$I = \frac{\left(\frac{\sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij} \left(X_i - \overline{X} \right) \left(X_j - \overline{X} \right)}{\sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij}} \right)}{\left(\frac{\sum_{i=1}^{n} \left(X_i - \overline{X} \right)^2}{n} \right)} \quad \# (1)$$

where:

- n is the total number of observations or spatial units
- w_{ij} is a weight indexing location of *i* relative to *j*
- X_i and X_j are observed values at locations *i* and *j*
- $\overline{X}$ is the mean of the variable X

In this report, the group uses a Queen Contiguity Weights Matrix, defining neighbors based on a segment or a point (a vertex). The remainder of the report will contain this matrix. Statisticians exploring spatial data often employ multiple weight matrices to assess their data. Matrices such as the queen contiguity weights matrix define spatial relationships critical for spatial regression models used later in this analysis. Different conclusions can be made on spatial relationships depending on the matrix used in one's study, which might capture spatial dependence in varying ways. Using carrying weight matrices, such as k-nearest neighbors, increases the validity of the results, ensuring the regression contains authentic spatial relationships.

Testing the Significance of Moran's I

To determine the significance of spatial autocorrelation, we conduct a hypothesis test where:

$$Null\ \ Hypothesis\ \ H_0 : No\ \ spatial\ \ autocorrelation\ \ \exists\ \left(Moran's\ \ I\ = \ 0\right)$$

$$Alternative\ \ Hypothesis\ \ H_a\ : Spatial\ \ autocorrelation\ \ is\ \ present\ \left(Moran's\ \ I\ \neq\ 0\right)$$

Significance is based on random permutations – values randomly assigned across locations multiple times (999 times). Each permutation generates a new Moran's I value to create a reference distribution representing the possible outcomes of values under the null hypothesis, H0. The observed Moran's I is then compared to the Moran's I values for the random permutations to address the significance. The pseudo-P-value, or the chance of getting a value as large as the observed value using samples from a distribution of Moran's I value for the actual variable and the 999 permutations sorted in descending order, is calculated by taking the rank of Moran's I and dividing it by 1,000. A significant Moran's I value indicates that clustering or dispersion is unlikely due to random chance.

Local Spatial Autocorrelation

Although Moran's I provides a global measure, Local Moran's I can assess clustering at specific locations. This method allows for granularity in the analysis. The local measure identifies significant clusters (high-high and low-low) and spatial outliers (high-low and low-high). High-high clusters demonstrate areas with high values surrounded by high values, whereas low-low clusters indicate areas of low values surrounded by low values. In contrast, spatial outliers are locations where values in one location are uniquely different than its neighbor. Observing these patterns across a region deepens the comprehension of local spatial relationships that a global analysis may not indicate.

Testing for local significance is like permutations in a global test -- repeated randomizing spatial data and recalculating the Local Moran's I statistics for each permutation generates a new distribution of values under the null hypothesis of spatial randomness. The observed Moran's I is compared to the new distribution to access significance. However, this distribution looks into the spatial data at a further refined scale. Identical to the global Moran's I, a pseudo-P-value is determined through the rank of the observed Local Moran's I to the permuted results. For example, if the Moran's I value is 0.68 and is higher than all 999 permutations, the p-value rank 0.0001 results in a 1 in 1000 chance of observing a Moran's I of 0.68 if no spatial autocorrelation is present.

OLS Regression and Assumptions

Ordinary Least Squares regression (OLS regression) is a standard for estimating the relationship between a dependent variable and multiple predictors. However, for an OLS regression to contain reliable and valid results, there are some assumptions that must be met that are listed below.

1. The relationship between the independent variables and the dependent variable should be linear and non-deterministic.
2. The model residuals, the differences between the observed and predicted values, should follow a normal distribution.
3. The observations in the dataset should also be independent of each other, meaning that one observation should not influence another.
4. The residuals should also be homoscedastic, or constant variance, across all predictor values. This means that the spread of residuals should not be dependent on or influenced by independent variables.
5. Finally, independent variables should not exhibit strong correlations with one another. Multicollinearity arises when two or more predictor variables are highly correlated, complicating the ability to determine their individual impacts on the outcome variable.

In spatial analyses, the independence assumption (item 4 in the list above) is often violated due to spatial autocorrelation, as geographical proximity in data typically demonstrates similar values. This coinciding relationship could introduce patterns in residuals that an OLS regression does not account for, indicating the need for spatial diagnostics. Moran's I is a standard diagnostic tool to supplement the need to quantify the extent of clustering in residuals. GeoDa computes Moran's I for the residuals, helping the team evaluate if spatial dependencies exist.

An additional method to identify spatial autocorrelation involves regressing residuals on their neighboring residuals, as defined by a spatial weights matrix. The team utilizes the Queen contiguity matrix in this analysis to specify neighborhood relations among residuals. A scatterplot of OLS residuals (OLS_RESIDU) against spatially lagged residuals (WT_RESIDU) visualizes spatial dependencies. The slope *b*, at the bottom of the scatterplot, quantifies the spatial correlation level. If *b* is statistically significant, residual spatial autocorrelation exists, indicating OLS limitations in the current spatial data.

GeoDa provides tools to assess OLS assumptions. Breusch-Pagan and Koenker-Bassett tests are for homoscedasticity. These two tests state a null hypothesis (H_0) that residuals are homoscedastic, and the alternative hypothesis (H_a) is the residuals are heteroscedastic. Examining the normality of residuals is applying the Jaque-Bera test, with a null hypothesis that residuals follow a normal distribution. In contrast, the alternative hypothesis is that residuals do not follow a normal distribution.

In addition to each of these tests, the F-statistic, R^2, and p-values of predictors assess an OLS model. The F-statistics test the null hypothesis that none of the predictors are significant. Failing to reject this hypothesis suggests that there is no predictive utility in the data. The significance of individual predictors in an OLS model is demonstrated by their p-values, whereas the R^2 and adjusted R^2 offer the fit or explanatory power of the model.

GeoDa's tools and Moran's I will be used to access the OLS's assumptions. If violations are indicated, spatial regression models can provide more accurate and robust estimates that address the spatial relationships demonstrated by the data.

Spatial Lag and Spatial Error Regression

GeoDa was used to complete the following regressions.

Spatial Lag Regression

A spatial lag is the variable's value at its neighboring spatial units. In a spatial lag regression model, spatial dependence is accounted for by including a spatially lagged dependent variable, calculated as a weighted average of neighboring observations, as an extra predictor. This approach uses the values of nearby areas to help predict the dependent variable, directly addressing potential spatial autocorrelation (e.g., when similar values cluster geographically). By doing so, the model enhances its accuracy when strong spatial relationships exist in the data.

$$LNMEDHVAL = \rho W \left(LNMEDHVAL \right) + \beta_0 + \beta_1 PCTVACANT + \beta_2 PCTSINGLES + \beta_3 PCTBACHMOR + \beta_4 LNNBELPOV100 + \varepsilon \quad (2)$$

where:

- $LNMEDHVAL$: the logarithmic-transformed Median Household Value
- $\rho W \left(LNMEDHVAL \right)$: *the spatial lag term*
  - o ρ: spatial autoregressive coefficient, which measures spatial dependence across neighboring areas
  - o W: spatial weights matrix
- β_0: the intercept term, which represents the value of the dependent variable when all the predictor variables are zero
- $PCTVACANT$: percentage of vacant houses
- β_1: coefficient for $PCTVACANT$
- $PCTSINGLES$: percentage of single-house units
- β_2: coefficient for $PCTSINGLES$
- $PCTBACHMOR$: percentage of individuals with a bachelor's degree or higher
- β_3: coefficient for $PCTBACHMOR$
- $LNNBELPOV100$: logarithmic-transformed number of households living in poverty
- β_4: coefficient for $LNNBELPOV100$
- ε: error term, where $\varepsilon \sim N\left(0, \sigma^2\right)$

Spatial Error Regression

Spatial error regression is an alternative approach for modeling spatial dependence. Unlike spatial lag models, it accounts for spatial autocorrelation through the error term rather than the dependent variable. This method assumes that spatial dependence stems from unobserved factors or random shocks in neighboring locations rather than direct interactions between nearby dependent variable values.

$$LNMEDHVAL = \beta_0 + \beta_1 PCTVACANT + \beta_2 PCTSINGLES$$
$$+ \ \beta_3 PCTBACHMOR \ + \beta_4 LNNBELPOV100 + \lambda W\varepsilon + u \# (3)$$

where:

- *LNMEDHVAL*: logarithmic-transformed Median Household Value
- β_0: intercept term, which represents the value of the dependent variable when all the predictor variables are zero
- *PCTVACANT*: percentage of vacant houses
- β_1: coefficient for *PCTVACANT*
- *PCTSINGLES:* percentage of single house units
- β_2: coefficient for *PCTSINGLES*
- *PCTBACHMOR:* percentage of individuals with a bachelor's degrees or higher
- β_3 : coefficient for *PCTBACHMOR*
- *LNNBELPOV* 100: logarithmic-transformed number of *households living in poverty*
- β_4 : coefficient for *LNNBELPOV* 100
- $\lambda W\varepsilon$: spatial lagged residuals
  - o λ: spatial autoregressive coefficient of errors
  - o W: spatial weights matrix
  - o ε: error term
- *u:* random noise

The assumptions needed for OLS regression (linearity, normality of residuals, homoscedasticity, no multicollinearity) are required for both spatial lag and spatial error regression models, except for the spatial independence of observations.

Spatial lag and spatial error regression models account for the data's spatial dependence, producing reliable and accurate estimates of the regression result. The spatial lag model addresses spatial dependence in the dependent variable by including a spatially lagged term of the dependent variable, accounting for the influence of neighboring observations. As the spatial influence is modeled, the residuals should ideally demonstrate that no spatial autocorrelation remains, meaning that all spatial patterns across the dependent variable are accounted for. Alternatively, the spatial error model accounts for spatial dependence through spatially structured error terms. After modeling the spatial structure, residuals consist solely of random noise (u) with no remaining spatial autocorrelation.

The results of the spatial lag regression and the spatial error regression will each be compared against the OLS regression. The Akaike Information Criterion (AIC) /Schwarz Criterion (SC), Log Likelihood, and the Likelihood Ratio Test will be applied to evaluate whether the spatial models perform better than OLS.

Model Comparison Criteria: AIC & SC

The Akaike Information Criterion (AIC) and Schwarz Criterion (SC) are measures of the goodness of fit of an estimated statistical mode, assessing the trade-off between the model's precision and complexity. It also quantifies the trade-offs between the model's fit of data and the number of parameters in favor of models with less complexity. The formula for both are as follows:

$$AIC = 2k - 2\ln(L) \# (4)$$

where

- k: number of parameters in the model,
- L: likelihood of the model, or the value of the likelihood function at the estimated parameters

and

$$SC = \ln(n)k - 2\ln(L) \# (5)$$

where

- n is the number of observations
- k is the number of parameters,
- L is the likelihood of the model.

The AIC and SC values provide little information but prove useful when the values of at least two models are compared. Smaller AIC and SC values indicate better fits.

Model Comparison Criteria: Log Likelihood

Log-likelihood is associated with the maximum likelihood estimation (MLE) method that estimates parameters to fit a statistical model to data. MLE selects model values that maximize the probability of observing the data. Log-likelihood, therefore, reflects the likelihood of the observed data based on the specified model parameters, where a higher log-likelihood value indicates a better model fit. In contrast, lower values reflect a poorer fit. Therefore, a less negative log-likelihood demonstrates a better model. Log-likelihood is only used to compare nested models, and OLS is a special case of both spatial lag and spatial error models.

Model Comparison Criteria: Likelihood Ratio Test

The likelihood ratio test is an additional test to compare nested models. The null hypothesis is the spatial lag (error) model is not a better specification than the OLS model. The alternative hypothesis is that the spatial lag (error) model better fits the OLS model. At a significance level of 5%, the null hypothesis would be rejected if the p-value is less than 0.05.

Model Comparison Criteria: Moran's I of Regression Residuals

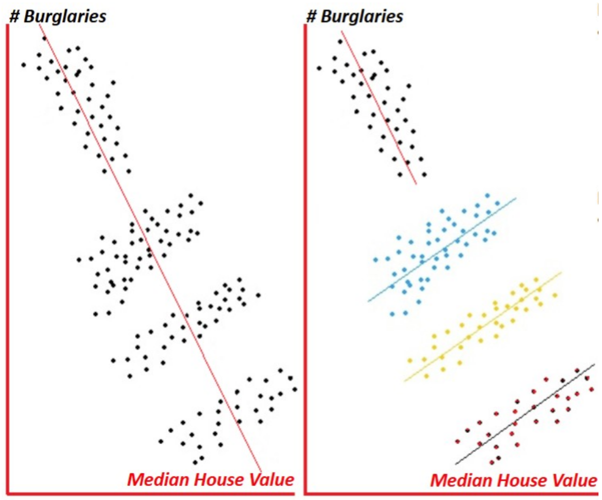
Comparing Moran's I of regression residuals is also a model comparison criterion. Moran's I value measures spatial autocorrelation, meaning it helps determine whether spatial dependence remains in the unexplained portion of the model after accounting for the predictors. A model with a lower Moran's I of the residuals, ideally close to zero, would indicate a better model that accounts for spatial dependence.

Geographically Weighted Regression (GWR)

Geographically weighted regression (GWR) analysis was conducted in R. This is a spatial regression technique to extend other regression models by enabling coefficients in the model to vary across space. This form of regression helps to address Simpson's Paradox, which occurs when trends observed across several groups of data either reverse or disappear when the data is combined. This paradox, pictured below, demonstrates the significance of also considering local spatial relationships during data analysis.

Figure 1

Simpson's Paradox



The equation for GWR is:

$$y_i = \beta_{i0} + \beta_{i1} x_{i1} + \beta_{i2} x_{i2} + \ldots + \beta_{\mathrm{im}} x_{\mathrm{im}} + \varepsilon_i = \beta_{i0} + \sum_{k=1}^{m} \beta_{ik} x_{ik} + \varepsilon_i \# (6)$$

where

- y_i: dependent variable (outcome) at location *i*
- x_{ik}: k^{th} predictor variable at location *i*
- β_{i0}: intercept term that varies by location *i*
- β_{ik}: represents the spatially varying coefficient for predictor k at location *i*
- ε_i: random error term at *i*

Geographically Weighted Regression (GWR) enables localized modeling by allowing relationships between dependent and independent variables to vary across space. For each location *i*, GWR fits a distinct ordinary least squares (OLS) regression with unique parameters—including an intercept, predictor coefficients, and an error term—capturing location-specific relationships between variables.

Since local regression requires multiple data points, GWR incorporates neighboring observations to estimate parameters for location *i*. Nearby locations are assigned higher weights in the regression, giving them greater influence on the results. This weighting is distance-based, meaning observations closer to *i* contribute more strongly to its parameter estimates than those farther away.

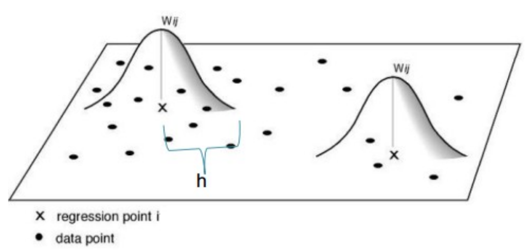
Bandwidths

In GWR, weights are adjusted on the type of bandwidth set. Bandwidth determines the weighting of nearby locations and can be either fixed or adaptive. A fixed bandwidth is when the number of observations vary around each point, *i*, but the distance (*h*) and thus the area remains constant. Distance *ij* is the distance between the regression point *i* and data point *j*. Points within the bandwidth receive weights that decrease with distance from i, while points outside the bandwidth are assigned a weight of zero. A possible way of defining the weights is:

$$w_{ij} = \begin{cases} e^{-0.5\left(\frac{distance_{ij}}{h}\right)^2}, if\ distance_{ij} \leq h \\ 0, otherwise \end{cases} \#(7)$$

Figure 2

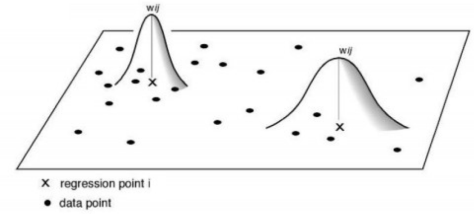
Fixed Bandwidth



In contrast, an adaptive bandwidth is where the number of observations will remain fixed, but the area will not be the same. Adaptive bandwidth kernel is appropriate when distribution varies across space (i.e., events are clustered or polygons are heterogeneously shaped or sized), whereas fixed bandwidth kernel will be more appropriate when the distribution of your observations is relatively stable across space (e.g., number of neighbors, size).

Figure 3

Adaptive Bandwidth



For an adaptive bandwidth, w_{ij}, can be defined as:

$$w_{ij} = ¿$$
¿

where h varies across observations.

In this report, an adaptive bandwidth was chosen for the regression model. An adaptive bandwidth is more appropriate for this problem than the fixed bandwidth, as the census block groups are heterogeneously shaped and sized, varying across space. Therefore, an adaptive bandwidth can better account for these variations.

Geographically Weighted Regression (GWR) retains the core OLS assumptions—linearity, normally distributed residuals, homoscedasticity, and no multicollinearity—while requiring at least 300 observations for reliable results. A key challenge in GWR arises when explanatory variables exhibit strong spatial clustering, leading to insufficient local variability, or when multiple variables show similar spatial patterns, creating local multicollinearity issues. These problems can be detected using the condition number, where values above 30, Null, or

-1.7976931348623158e+308 indicate unreliable results that should be interpreted cautiously. Unlike OLS, GWR does not provide p-values due to the impractical number of statistical tests needed to evaluate location-specific coefficients and standard errors, though it does output these parameters for each regression point along with global standard errors. Thus, while GWR captures spatial heterogeneity effectively, proper diagnosis of local multicollinearity and spatial clustering is essential for valid interpretation.

Results

Spatial Autocorrelation

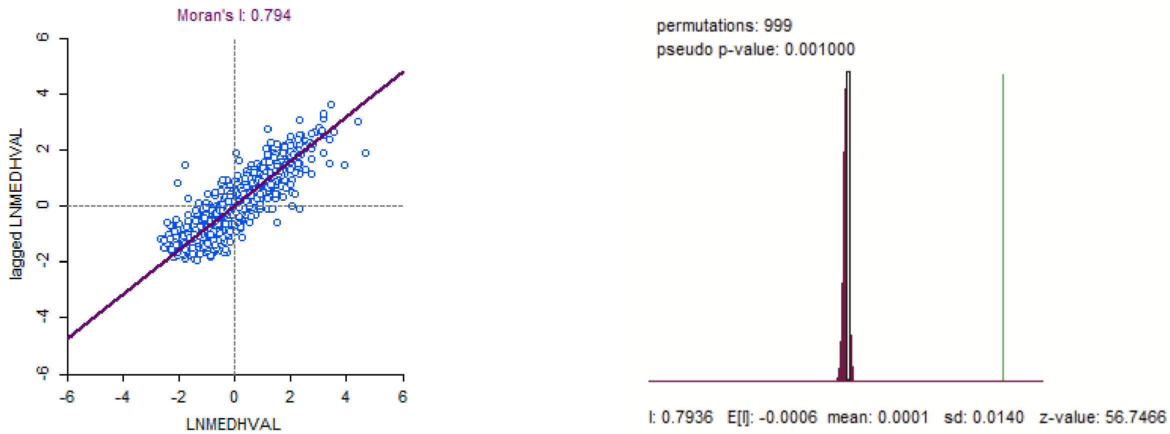
The group calculated the Global Moran's I and conducted random permutations to test the null hypothesis that no spatial autocorrelation is present when determining if the log-transformed median household value (LNMEDHVAL) exhibits spatial autocorrelation across Philadelphia's block groups.

Global Moran's I

The Global Moran's I for the log-transformed median household value returned to 0.794 (positive), suggesting a high level of spatial clustering. A p-value of 0.001000 from 999 random permutations rejects the null hypothesis, indicating that log-transformed median household value is spatially autocorrelated across the study area.

Figure 4

Moran's Scatterplot of Lagged LNMEDHVAL against LNMEDHVAL & 999 Permutations



Local Spatial Autocorrelation

To examine localized spatial patterns in the log-transformed median household value, we applied Local Moran's I to create significance and cluster maps. These demonstrate areas that may or may not exhibit significant clustering or outlier behavior.

Figure 5

LISA Significance Map

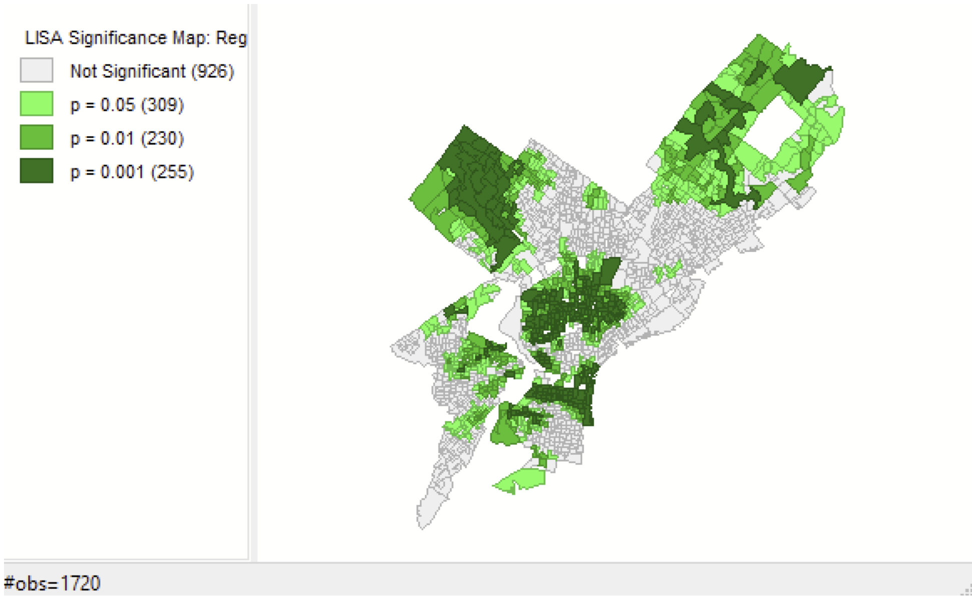
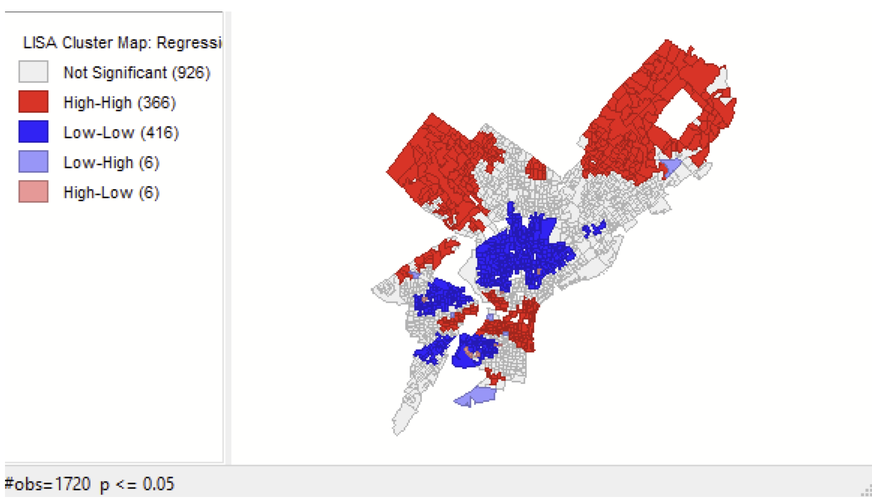


Figure 6

LISA Cluster Map



The following spatial patterns can be observed from our maps:

- **High-High Clusters**:
    - Neighborhoods like Center City and Chestnut Hill, where high LNMEDHVAL values cluster together, are known for higher housing values and wealthier demographics.
- **Low-Low Clusters**:
    - Mainly North and Southwest Philadelphia, exhibiting clustering of lower LNMEDHVAL values, these neighborhoods lower housing values and often higher poverty rates.
- **High-Low and Low-High Outliers**:
    - High-low outliers (high-value neighborhoods surrounded by low-value neighborhoods) appear at the edges of gentrifying neighborhoods or regions undergoing transition. In contrast, low-high areas (low-value neighborhoods surrounded by higher-value neighborhoods) are typically adjacent to wealthier areas in West and South Philadelphia. These outliers could reflect localized shifts in housing value in a neighborhood anchored by an academic or employing institution, such as a university or hospital, or good transportation access along Interstate 76.

Results of OLS Regression

Table 1

Output of OLS Regression

```
REGRESSION
----------
SUMMARY OF OUTPUT: ORDINARY LEAST SQUARES ESTIMATION
Data set            :  RegressionData
Dependent Variable  :   LNMEDHVAL  Number of Observations: 1720
Mean dependent var  :      10.882  Number of Variables   :    5
S.D. dependent var  :     0.62972  Degrees of Freedom    : 1715

R-squared           :    0.662300  F-statistic           :     840.869
Adjusted R-squared  :    0.661513  Prob(F-statistic)     :           0
Sum squared residual:     230.332  Log likelihood        :    -711.493
Sigma-square        :    0.134304  Akaike info criterion :     1432.99
S.E. of regression  :    0.366475  Schwarz criterion     :     1460.24
Sigma-square ML     :    0.133914
S.E of regression ML:    0.365942

-----------------------------------------------------------------------------
       Variable      Coefficient      Std.Error     t-Statistic   Probability
-----------------------------------------------------------------------------
          CONSTANT       11.1138      0.0465318        238.843      0.00000
         LNNBELPOV    -0.0789035      0.0084567        -9.3303      0.00000
        PCTBACHMOR     0.0209095    0.000543184        38.4944      0.00000
        PCTSINGLES    0.00297695    0.000703155        4.23371      0.00002
         PCTVACANT    -0.0191563    0.000977851       -19.5902      0.00000
-----------------------------------------------------------------------------

REGRESSION DIAGNOSTICS
MULTICOLLINEARITY CONDITION NUMBER   12.990609
TEST ON NORMALITY OF ERRORS
TEST                  DF           VALUE             PROB
Jarque-Bera            2           778.9646          0.00000

DIAGNOSTICS FOR HETEROSKEDASTICITY
RANDOM COEFFICIENTS
TEST                  DF           VALUE             PROB
Breusch-Pagan test     4           162.9108          0.00000
Koenker-Bassett test   4            61.6992          0.00000
SPECIFICATION ROBUST TEST
TEST                  DF           VALUE             PROB
White                 14           111.3224          0.00000

DIAGNOSTICS FOR SPATIAL DEPENDENCE
FOR WEIGHT MATRIX : RegressionData
   (row-standardized weights)
TEST                           MI/DF        VALUE           PROB
Moran's I (error)              0.3129        22.3664         0.00000
Lagrange Multiplier (lag)        1          930.1626         0.00000
Robust LM (lag)                  1          441.1061         0.00000
Lagrange Multiplier (error)      1          490.5691         0.00000
Robust LM (error)                1            1.5126         0.21875
Lagrange Multiplier (SARMA)      2          931.6751         0.00000
```

The percentage of vacant houses, the percentage of single-house units, the percentage of individuals with a bachelor’s degree, and the number of households in poverty are all significant in predicting median house values in Philadelphia, with each predictor reaching a high level of statistical significance ($p < 0.0001$). This model accounts for roughly 66% of the variance in logarithmic-transformed median house values (adjusted $R^2 = 0.662$), highlighting that the selected neighborhood characteristics in the analysis are a potential indicator of home values. The model's F-statistic is highly significant ($p < 0.001$), demonstrating the strength of the model's predictors.

Additional Diagnostic Testing

To further assess model validity, we examined OLS assumptions:

Multicollinearity. A Condition Number of 12.99 suggests multicollinearity is within acceptable limits, meaning the model characterizes predictor effects reliably.

Normality of Residuals. A Jarque-Bera test statistic of 778.96 and a probability less than 0.00001 demonstrate non-normality in residuals, which may skew inference and warrant careful interpretation of statistical significance.

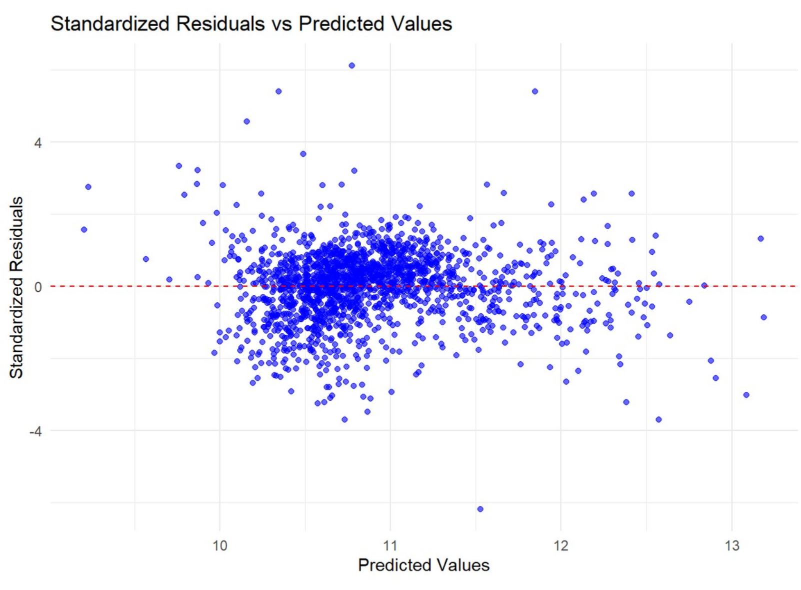
Homoscedasticity. The Breusch-Pagan and Koenker-Bassett tests, returning 162.91 and 61.70, respectively, with a probability less than 0.00001, reveal significant heteroscedasticity. Therefore, the residual variance is not constant. A lack of homoscedasticity hinders how reliable our confidence intervals and standard error estimates are, resulting in a potential bias in predictor effects.

Although LNMEDHVAL is significantly associated with each predictor in the model, spatial dependencies and heteroscedasticity affect the OLS model's reliability. Moran's I and Local Moran's I show that spatial autocorrelation is a limitation of OLS in spatial data, resulting in the need for spatial lag or error models. Local Moran's I identified high-value clusters in affluent parts of Philadelphia like Center City and low-value clusters in economically disadvantaged neighborhoods such as North Philadelphia, demonstrating the necessity for the model to be improved if it were an OLS.

The Breusch-Pagan and Koenker-Bassett tests both returned significant results, indicating the presence of heteroscedasticity. This signifies residual variance is not constant across predictor values and could alter the reliability of confidence intervals and standard errors in the model. The results of these tests are consistent and indicate heteroscedasticity in the OLS model. However, this conclusion is different than that from our residual-by-predicted plot in the first paper, where the report concluded heteroscedasticity was not present.

Figure 7

Scatterplot of Standardized Residuals vs. Predicted Values (from previous report)

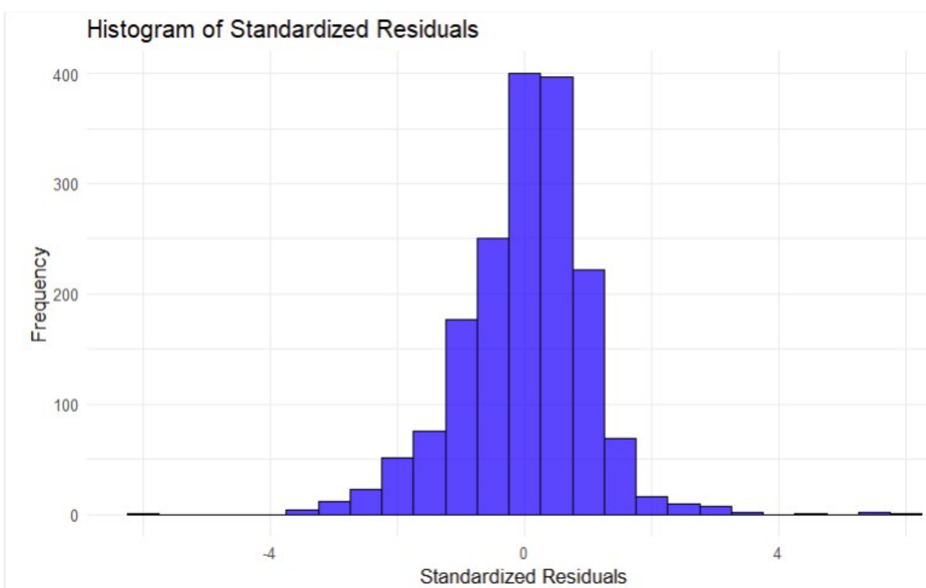


The apparent inconsistency may arise because the residuals in Homework 1 did not exhibit strong heteroscedasticity overall, yet they showed slightly greater variability as predicted values increased. While this pattern was not visually pronounced, formal statistical tests detected subtle heteroscedasticity in certain ranges of predicted values. Thus, even though graphical diagnostics initially suggested minimal heteroscedasticity, the significant test results indicate that caution is warranted when interpreting the model's estimates.

Similarly, the Jarque-Bera test (JB = 778.96, $p < 0.00001$) indicated a significant deviation from normality in the residuals, potentially affecting the reliability of confidence intervals and hypothesis tests. This finding contrasts with the histogram from Homework 1, which suggested a roughly normal distribution. The discrepancy likely stems from the Jarque-Bera test's high sensitivity to even minor departures from normality.

Figure 8

Histogram of Standardized Residuals (from previous report)

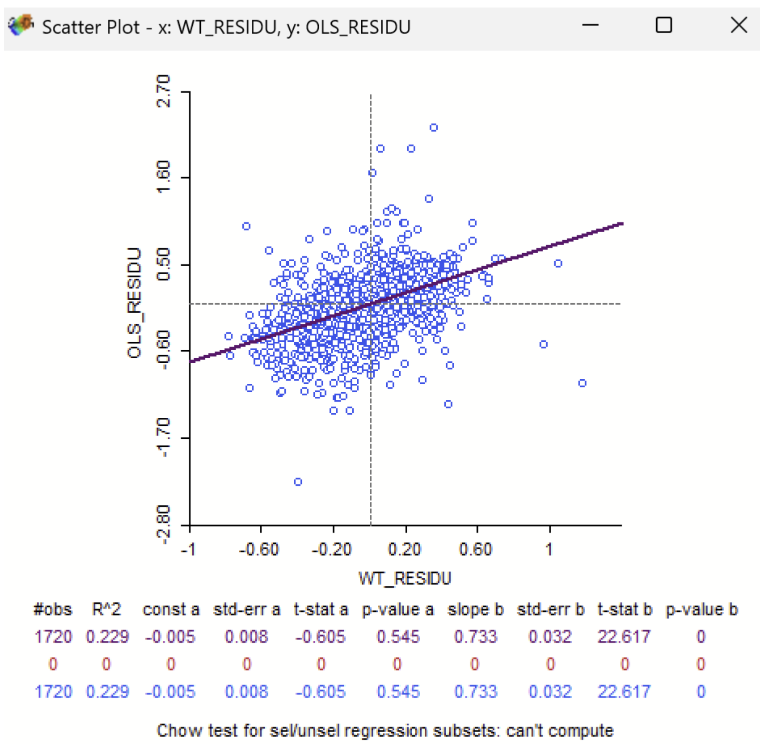


Spatial Dependence Analysis

A scatterplot of OLS residuals (OLS_RESIDU) against spatially lagged residuals (WT_RESIDU), including statistics on spatial dependencies, is shown below.

Figure 9

Scatterplot of OLS_RESIDU Against WT_RESIDU

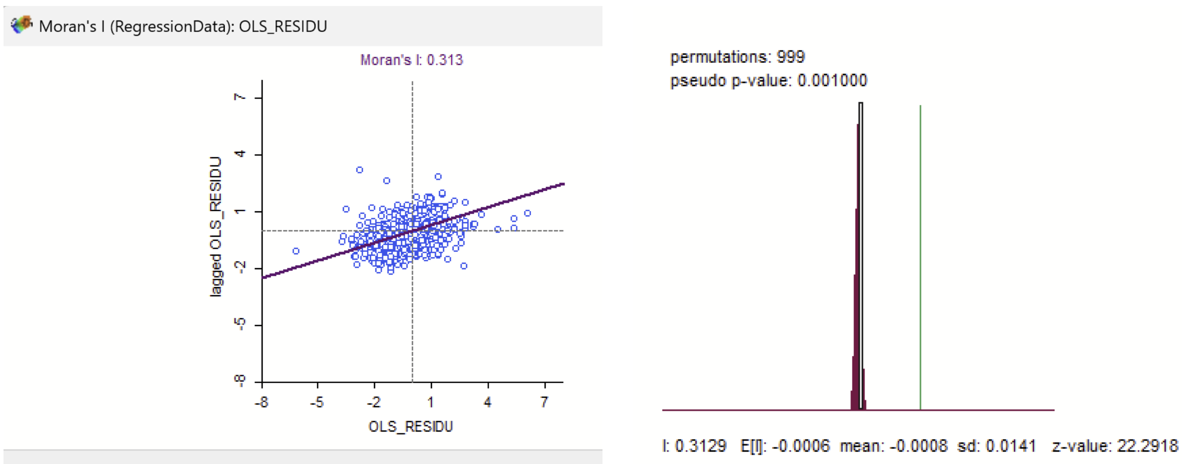


The slope b, with a value of 0.733, has a p-value of 0, indicating that spatial autocorrelation is significant.

A Moran's I was also used to test for spatial autocorrelation in the residuals, resulting in a positive Moran's I value of 0.313 ($p < 0.00001$). This confirms the presence of spatial autocorrelation and indicates spatial dependencies are unaddressed in the previous OLS model residuals.

Figure 10

Moran's I scatterplot and 999 Permutations Histogram – OLS Model



Spatial Lag and Spatial Error Regression Results

Table 2

Spatial Lag Regression Model Results

```
>>04/03/25 09:44:35
REGRESSION
----------
SUMMARY OF OUTPUT: SPATIAL LAG MODEL - MAXIMUM LIKELIHOOD ESTIMATION
Data set            : RegressionData
Spatial Weight      : RegressionData
Dependent Variable  :   LNMEDHVAL  Number of Observations: 1720
Mean dependent var  :      10.882  Number of Variables   :    6
S.D. dependent var  :     0.62972  Degrees of Freedom    : 1714
Lag coeff.   (Rho)  :    0.651097

R-squared           :    0.818564  Log likelihood        :     -255.74
Sq. Correlation     : -            Akaike info criterion :      523.48
Sigma-square        :    0.071948  Schwarz criterion     :      556.18
S.E of regression   :    0.268231

-----------------------------------------------------------------------------
       Variable       Coefficient      Std.Error       z-value    Probability
-----------------------------------------------------------------------------
       W_LNMEDHVAL      0.651097       0.0180501        36.0716     0.00000
          CONSTANT       3.89846        0.201114        19.3843     0.00000
         LNNBELPOV    -0.0340547      0.00629287       -5.41163     0.00000
         PCTVACANT    -0.0085294     0.000743667       -11.4694     0.00000
        PCTSINGLES    0.00203342      0.00051577         3.9425     0.00008
        PCTBACHMOR    0.00851381     0.000521935         16.312     0.00000
-----------------------------------------------------------------------------

REGRESSION DIAGNOSTICS
DIAGNOSTICS FOR HETEROSKEDASTICITY
RANDOM COEFFICIENTS
TEST                                     DF      VALUE         PROB
Breusch-Pagan test                       4       220.3884      0.00000

DIAGNOSTICS FOR SPATIAL DEPENDENCE
SPATIAL LAG DEPENDENCE FOR WEIGHT MATRIX : RegressionData
TEST                                     DF      VALUE         PROB
Likelihood Ratio Test                    1       911.5067      0.00000
```

The W_LNMEDVHAL term has a coefficient of 0.651097 and a p-value of 0.00000, resulting in a statistically significant coefficient. This means that the median house value in the area is positively associated with the median house values in neighboring areas. Notably, a one-unit increase in the spatially weighted average of median house values in neighboring areas increases the median house value in the target area by 0.6511 units, holding all other predictors constant. This shows that house values in nearby neighborhoods will likely influence the house value in the target area.

Each predictor is statistically significant. Compared to the coefficients from the OLS regression, all coefficients in this model are smaller. LNNBELPOV has a lower coefficient, shifting from -0.0789 to -0.034; the PCTBACHMOR coefficient shifted from 0.0209 to 0.0085; the PCTSINGLES coefficient shifted from 0.0030 to 0.0020, and PCTVACANT from -0.019 to -0.008. This observed reduction shows that some of the associations between the predictors and house values in the OLS model were caused by spatial dependence. Adjusting for spatial dependence in the spatial lag model, the beta coefficients cannot be interpreted in the same way as any other OLS since the total effects of the predictors include both direct and indirect effects.

The Breusch-Pagan test, a figure related to heteroscedasticity, returned a value of 220.3884, which is high compared to typical values. Additionally, this value is significant, with a p-value of 0.00000. This indicates that although spatial dependence in the model is accounted for by including lagged variables in an additional predictor, heteroscedasticity is still affecting the residuals.

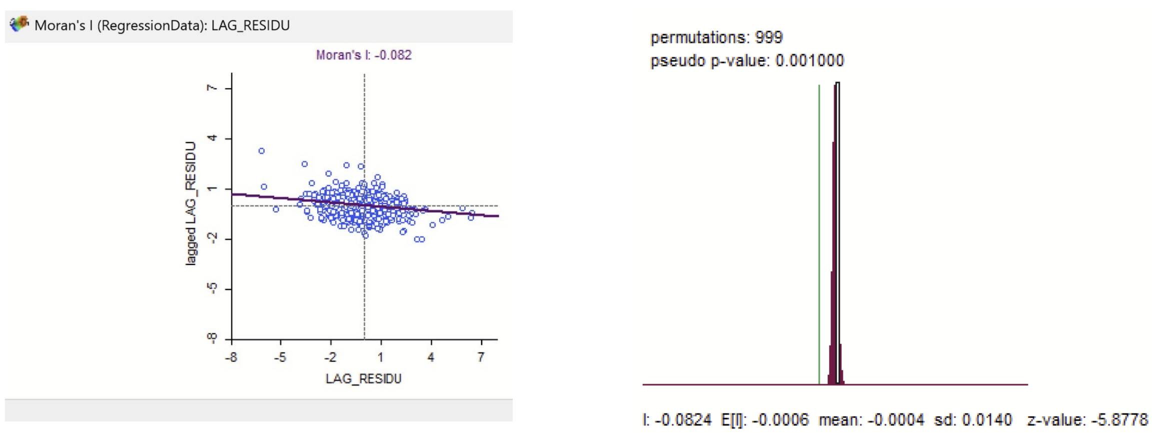
Comparison with the OLS Regression Model

The spatial lag model has an Akaike Information Criterion (AIC) of 523.48 and a Schwarz Criterion (SC) of 556.18. The OLS model has an AIC of 1432.99 and an SC of 1460.24. The spatial lag model has much lower AIC and SC values in comparison, suggesting that this fits the data better and captures spatial dependencies that the OLS regressions could not. A lower SC value demonstrates that increased complexity in the model resulted in an improved model fit compared to OLS.

The log-likelihood of the OLS regression is -711.493, which is less than the spatial lag model at -255.74. A less negative log-likelihood indicates a better-fitting model than the spatial lag model. Finally, the likelihood ratio test for the spatial lag model produced a statistically significant value of 911.5067, indicating the model provides a better fit to the data than the OLS model. This substantial improvement demonstrates that the spatial lag model captures a more significant variance than the OLS model failed to achieve, likely due to spatial dependence within the given data.

Figure 11

Moran's I scatterplot and 999 Permutations Histogram - Spatial Lag Model



A Moran's I value of -0.083 compared to 0.313 from the OLS model indicates less spatial autocorrelation in these residuals than in the OLS residuals.

In sum, the spatial lag model performs better than the OLS regression based on each of the criteria discussed previously. Although heteroscedasticity exists, the spatial lag model accounts for some spatial dependency that the previous OLS regression could not account for.

Spatial Error Regression

Table 3

Spatial Error Regression Model Results

```
>>04/03/25 09:51:51
REGRESSION
----------
SUMMARY OF OUTPUT: SPATIAL ERROR MODEL - MAXIMUM LIKELIHOOD ESTIMATION
Data set             : RegressionData
Spatial Weight       : RegressionData
Dependent Variable   :   LNMEDHVAL  Number of Observations: 1720
Mean dependent var   :   10.882000  Number of Variables   :    5
S.D. dependent var   :    0.629720  Degrees of Freedom    : 1715
Lag coeff. (Lambda)  :    0.814918

R-squared            :    0.806957  R-squared (BUSE)      : -
Sq. Correlation      : -            Log likelihood        : -372.690368
Sigma-square         :   0.0765508  Akaike info criterion :     755.381
S.E of regression    :    0.276678  Schwarz criterion     :     782.631

-----------------------------------------------------------------------------
       Variable       Coefficient     Std.Error       z-value    Probability
-----------------------------------------------------------------------------
          CONSTANT       10.9064       0.0534678       203.981      0.00000
         LNNBELPOV    -0.0345341      0.00708933      -4.87127      0.00000
        PCTBACHMOR    0.00981293     0.000728964       13.4615      0.00000
        PCTSINGLES    0.00267792     0.000620832       4.31343      0.00002
         PCTVACANT   -0.00578308     0.000886701      -6.52201      0.00000
            LAMBDA      0.814918        0.016373       49.7719      0.00000
-----------------------------------------------------------------------------

REGRESSION DIAGNOSTICS
DIAGNOSTICS FOR HETEROSKEDASTICITY
RANDOM COEFFICIENTS
TEST                                    DF      VALUE        PROB
Breusch-Pagan test                      4       210.9923     0.00000

DIAGNOSTICS FOR SPATIAL DEPENDENCE
SPATIAL ERROR DEPENDENCE FOR WEIGHT MATRIX : RegressionData
TEST                                    DF      VALUE        PROB
Likelihood Ratio Test                   1       677.6059     0.00000
```

The λ (labeled LAMBDA in Table 3) in the spatial regression is 0.814918, statistically significant with a p-value of 0.00000. This demonstrates strong positive spatial autocorrelation in the error terms, meaning that unobserved factors influencing the dependent variable exhibit high degrees of spatial clustering.

Each predictor (LNNBELPOV, PCTBACHMOR, PCTSINGLES, and PCTVACANT) is statistically significant. Compared to each predictor coefficient in the original OLS model, all coefficients in the spatial error model are smaller. LNNBELPOV decreased from -0.0789 to -0.034, PCTBACHMOR decreased from 0.021 to 0.0098, PCTSINGLES decreased from 0.0030 to 0.0026, and PCTVACANT decreased from -0.019 to -0.005. This indicates that some associations between the predictors and the dependent values from the OLS were due to spatial dependence. Therefore, the coefficients are still smaller but statistically significant after the spatial error model absorbs some spatial dependence through a spatial error term. A Breusch-Pagan test produces a statistically significant (p-value of 0.00000) value of 210.9923. Despite accounting for spatial dependence in the model by including a spatial error term, heteroscedasticity affects the residuals.

Comparison with OLS Regression Model

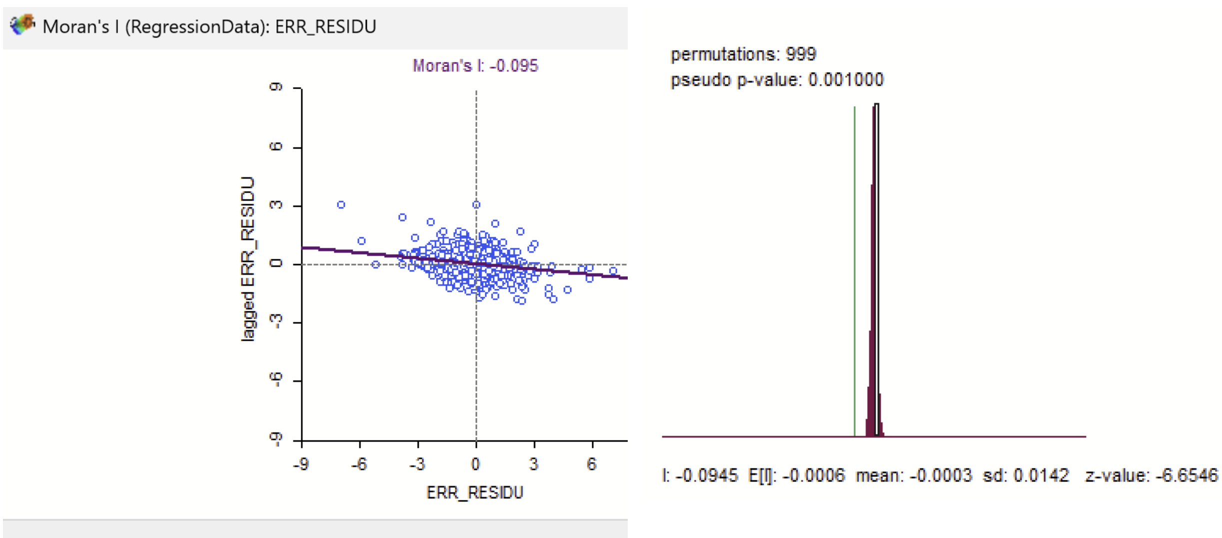
In the spatial error model, the AIC was 755.381, and the SC was 782.631, whereas the AIC and SC of the OLS model were 1432.99 and 1460.24, respectively.

Similar to the spatial lag model, the spatial error model resulted in lower AIC and SC values than the OLS model, suggesting that the spatial error model is a much better fit for the data, capturing actual dependencies in the data that the OLS could not. A lower SC value indicates that the model had a better fit, even with a more complex model.

The log-likelihood of the spatial error model (-372.690) is much greater than the OLS regression (-711.493), indicating that the model is a better fit. Additionally, the likelihood ratio test of the spatial error model produced a considerable, statistically significant value of 677.6059, which indicates that the spatial error model fits the data significantly better than the OLS model and captures more variance.

Figure 12

Moran's I Scatterplot and Histogram of 999 Permutations for Spatial Error Model



The spatial error model resulted in a smaller Moran's I value than the OLS model, at -0.095 (compared to 0.313), demonstrating a reduction in spatial autocorrelation in the residuals compared to the OLS model.

Based on these criteria, the spatial lag model performed better than the OLS regression. Even if the model's residuals are heteroscedastic, this model still accounts for some spatial dependency, improving the overall model fit.

Spatial Lag vs. Spatial Error

The AIC and SC values (523.48 and 556.18, respectively) for the spatial lag model are lower than those for the spatial error model (755.381 and 782.631, respectively), indicating that the spatial lag model performs better. When considering model complexity, the spatial lag model proves superior to the spatial error model in accounting for spatial dependence, leading to enhanced model performance.

Geographically Weighted Regression Results

Table 4

Global Weighted Regression Results

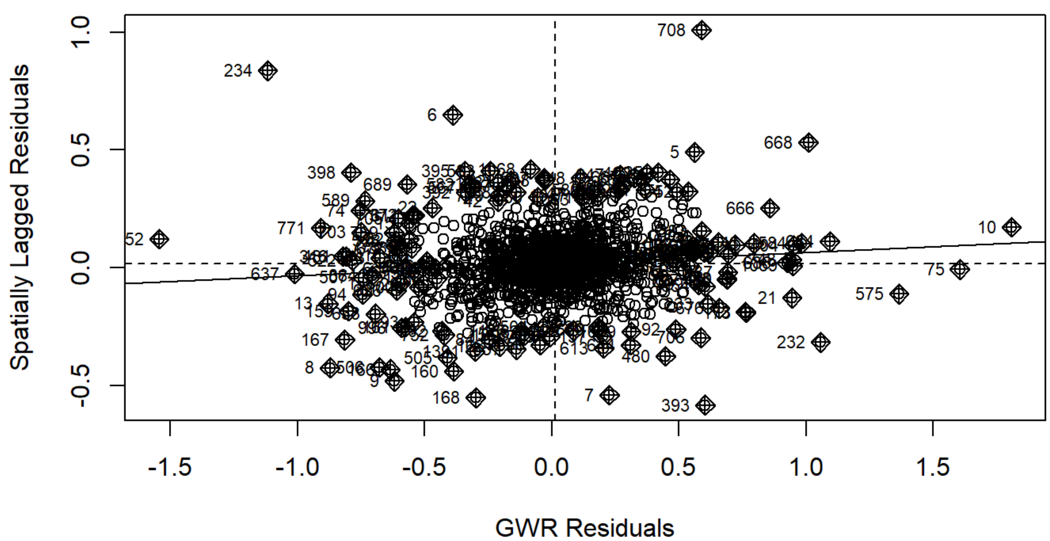
```
Summary of GWR coefficient estimates at data points:
                   Min.    1st Qu.     Median    3rd Qu.       Max.  Global
X.Intercept.  9.6727618 10.7143173 10.9542384 11.1742009 12.0831381 11.1138
LNNBELPOV    -0.2365244 -0.0733572 -0.0401186 -0.0126657  0.0948768 -0.0789
PCTBACHMOR    0.0010974  0.0101380  0.0149279  0.0202187  0.0347258  0.0209
PCTSINGLES   -0.0249706 -0.0075550 -0.0016626  0.0042280  0.0143340  0.0030
PCTVACANT    -0.0317407 -0.0142383 -0.0089599 -0.0035770  0.0167916 -0.0192
Number of data points: 1720
Effective number of parameters (residual: 2traceS - traceS'S): 360.5225
Effective degrees of freedom (residual: 2traceS - traceS'S): 1359.477
Sigma (residual: 2traceS - traceS'S): 0.2762201
Effective number of parameters (model: traceS): 257.9061
Effective degrees of freedom (model: traceS): 1462.094
Sigma (model: traceS): 0.2663506
Sigma (ML): 0.245571
AICc (GWR p. 61, eq 2.33; p. 96, eq. 4.21): 660.7924
AIC (GWR p. 96, eq. 4.22): 308.7123
Residual sum of squares: 103.7248
Quasi-global R2: 0.8479244
```

The global R^2 of the geographically weighted regression (GWR) is 0.6913, which is higher than the R^2 value of 0.662 from the OLS regression. This indicates that the GWR model explains 69.13% of the variance in the logarithmically transformed median house value, demonstrating this model’s ability to better capture the variability in the data compared to an OLS model.

The GWR model's AIC is 1282.71, and the AICc (Akaike Information Criterion corrected) is 1282.776. Compared to the OLS, spatial lag, and spatial error models, which have AIC values of 1432.99, 523.48, and 755.381, respectively, the GWR has the lowest/highest AIC values, indicating that this model has the best fit.

Figure 13

Moran’s Scatterplot of GWR Residuals

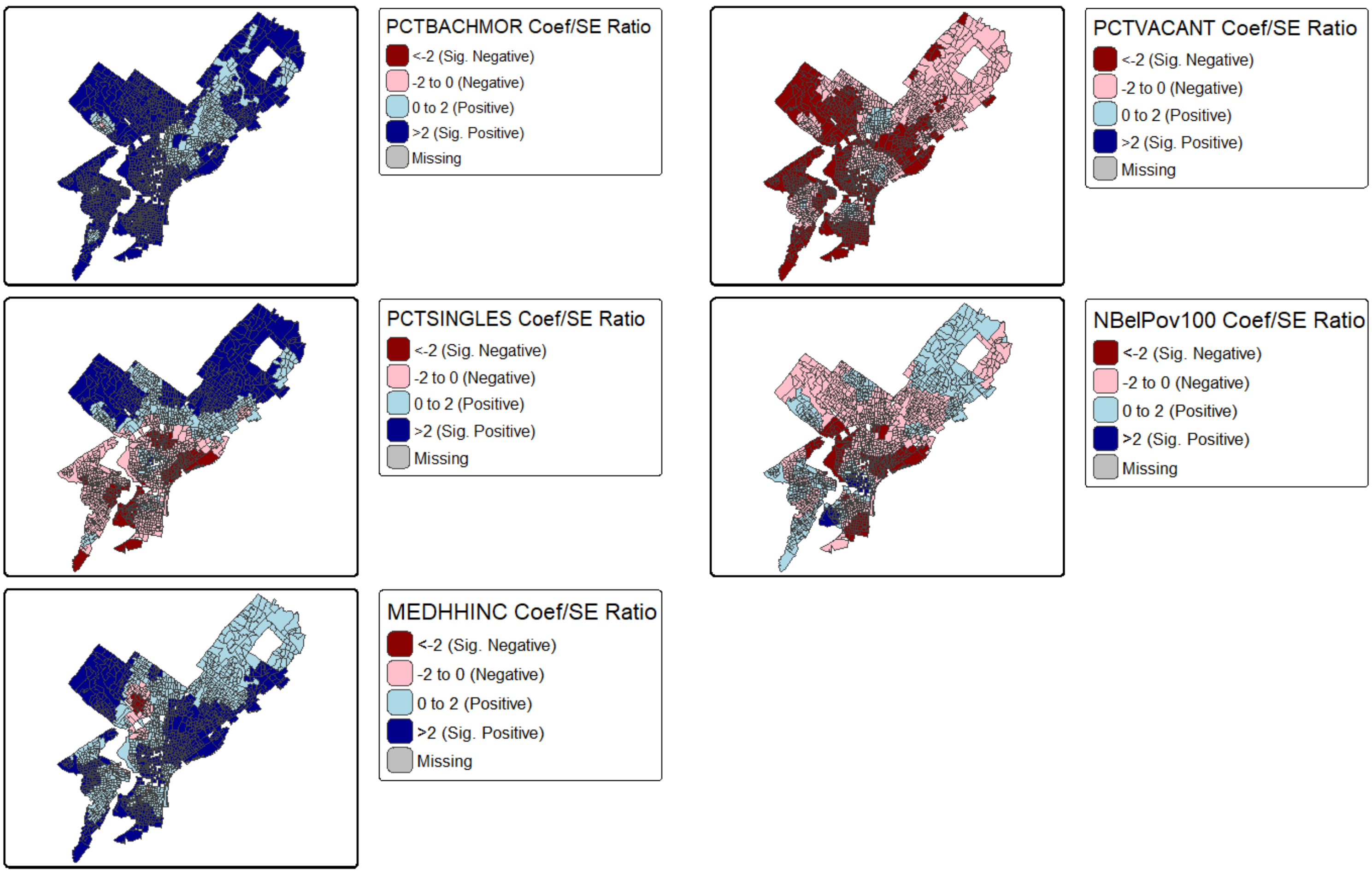


In the GWR model, there is even less spatial autocorrelation demonstrated by Moran’s I value of 0.0334 compared to the OLS residuals with a value of 0.313. GWR’s Moran’s I value is smaller than the absolute value of the spatial lag and spatial error models at 0.083 and 0.095, respectively.

Maps of the coefficients divided by their standards errors from the local regression results indicate a possible significance in the relationships between predictors and the dependent variable. Standardized coefficients below -2 or above 2 (dark red and dark blue) suggest that the relationships may be statistically significant.

Figure 14

Local Regression Results



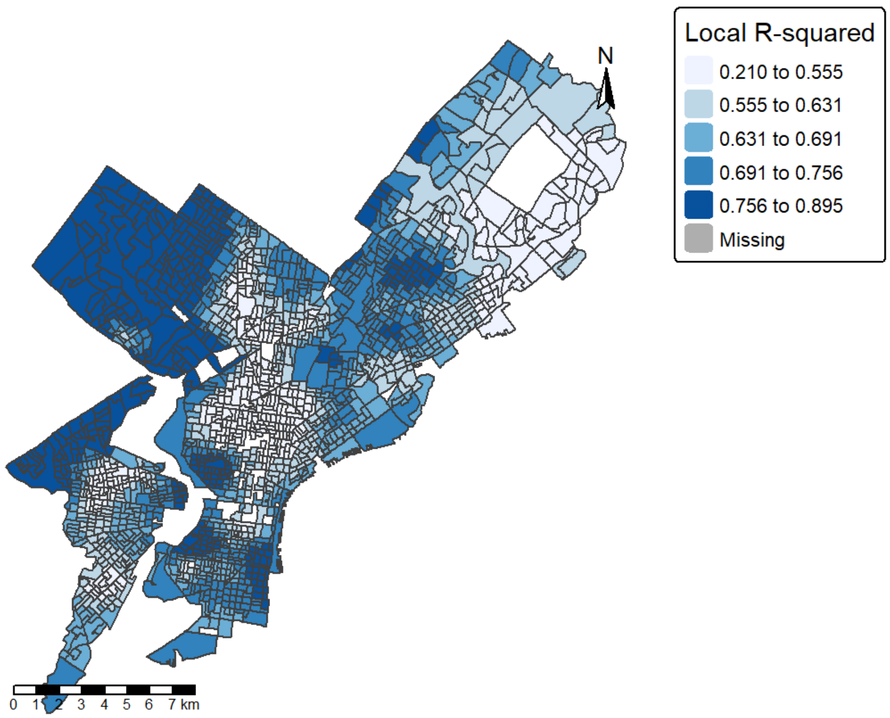
These maps provide compelling geospatial heterogeneity in the relationships between the predictors and the logarithmic-transformed median house values.

- LNNBELPOV (Log-Transformed Percentage Below Poverty):
    - Negative relationships with high-magnitude standardized coefficients are observed in West Philadelphia and southern areas along the Delaware River, suggesting these relationships are likely significant.
- Percentage with Bachelor's Degrees (PCTBACH):
    - Across all of Philadelphia, there is a positive relationship between this higher education level and median house values. Positive standardized coefficients larger than two are observed in both West Philadelphia and South Philadelphia, highlighting strong positive relationships with house values in these regions.
- Percentage of Single Housing Units (PCTSINGLES):
    - Standardized coefficients greater than 2 are evident in West Philadelphia, indicating significant positive relationships with median house values in this area. Negative relationships with median house values are present in the central and southern areas, where this is likely significant in some central regions.
- Percentage of Vacant Housing Units (PCTVACANT):
    - Negative standardized coefficients are found in pockets of the central, western, and southern regions, suggesting that vacancy has a negative association with median house values in these areas.

Additionally, a choropleth map of local R^2 values was produced to support insight into how the model fits regionally throughout Philadelphia County.

Figure 15

Choropleth Map of Local R^2 in Philadelphia County



Local R^2 is the highest in the northwestern and southeastern corners of Philadelphia and lowest around Center City and Far Northeast (adjacent to Bucks County, PA). The model can account for the most significant amount of variance in the dependent variable in the outer regions of Philadelphia, especially in West Philadelphia and Northwest Philadelphia, where the regression model can account for >60% of the variance in the logarithmic-transformed median house values. It accounts for the least variance around Center City and Far Northeast, where the model cannot predict median house values.

Discussion and Limitations

This report examined Spatial Lag, Spatial Error, and Geographically Weighted Regression (GWR) models to quantify the relationship between neighborhood characteristics and median house values across Philadelphia County. The results demonstrated that AIC, SC, and Moran's I were all lower compared to the OLS model, confirming that the spatial autocorrelation of residuals decreased. Ultimately, the model's reliability from the first report on OLS regression improved.

Table 5

Comparison of AIC, SC, Moran's I, and Log Likelihood Across All Models

Metric	OLS	Spatial Lag	Spatial Error	GWR
AIC	1,433	523	755	309
SC	1,460	556	783	
Moran's I	0.313	-0.082	-0.095	0.033
Log Likelihood	-711.49	-255.74		

Comparing the Spatial Lag and the Spatial Error model, the Spatial Lag model was better than the Spatial Error model based on AIC, SC, and Moran's I. The GWR model performed the best among the three models, with an AIC of 309 and a Moran's I of 0.033. There are local variations in the relationships between predictors and the dependent variable, which global models, such as an OLS regression, mask.

Even when accounting for spatial autocorrelation, the significance of the existing independent variables was still maintained. However, since spatial autocorrelation is present and is ignored in the OLS model, we can say that the significance of the β coefficients, which may have been overestimated, was corrected to be more accurate.

However, each of the models still has limitations. Homoscedasticity was assumed in the Spatial Lag, Spatial Error, and GWR models; each was heteroscedastic according to the Breusch-Pagan test. Second, we assumed spatial stationarity, which means the modeled relationships are constant across space in the Spatial Lag and Spatial Error models, unlike in GWR.

The spatial lag and spatial error models aim to address the presence of spatial dependence, but there is little difference between them. The Spatial Lag Model adds a spatially lagged dependent variable as an additional predictor in the regression, assuming the values of the dependent variable in one location are directly influenced by the values of the dependent variable in neighboring locations. The Spatial Error Model addresses spatial dependence in the residuals by including a spatially structured error term, that spatial autocorrelation in residuals are omitted, unobserved spatial variables. The spatial error term accounts for this dependence.

The key distinction between these models lies in the source of spatial dependence they account for. The spatial lag model operates under the assumption that the dependent variable at a given location is directly influenced by the values of that same variable in neighboring locations. In contrast, the spatial error model treats spatial dependence as a product of unobserved, spatially autocorrelated factors that manifest through the error term rather than directly affecting the dependent variable.

It is worth noting that ArcGIS was not used to perform geographically weighted regression (GWR) in this analysis due to certain methodological constraints. While ArcGIS employs AICc (a small-sample-corrected version of Akaike's Information Criterion) for bandwidth selection, this approach may introduce inconsistencies when comparing results with models that rely on standard AIC. Additionally, the bandwidth optimization algorithm in ArcGIS Pro does not always guarantee convergence to the globally optimal AICc value, which could lead to less reliable model fits compared to alternative implementations.